Higher-Dimensional Local Euler Characteristic Computation via Dual Cell Valuations on N-Dimensional Lattices

Formal Notes on Digital Topology and Valuation Theory

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Abstract

We extend the dual-decomposition valuation framework for the Euler characteristic from 2-dimensional regular tilings to arbitrary N-dimensional uniform and regular cell complexes. By decomposing the global Euler-Poincaré formula across dual Voronoi cells centered at primal vertices, we prove that the topological Euler characteristic of an N-dimensional binary digital image (represented by active voxels or N-cells) is strictly locally computable. We provide the complete combinatorial formulation for hypercubic lattices $\mathbb{Z}^N$, provide explicit local formulas for 3-dimensional volume processing on $\mathbb{Z}^3$ ($2 \times 2 \times 2$ bit octocubes), and establish the connection to discrete Morse theory and spherical link valuations.

1 N-Dimensional Geometric and Topological Framework

Let $\mathbb{M}^N$ be an N-dimensional simply connected Riemannian manifold of constant curvature ($\mathbb{R}^N$, $\mathbb{S}^N$, or $\mathbb{H}^N$).

Definition 1.1 (N-Dimensional Regular/Uniform Honeycomb)**.** Let $\mathcal{T}$ be a locally finite, regular or uniform cellular decomposition of $\mathbb{M}^N$. For each dimension $k \in \{0, 1, \ldots, N\}$, let $\mathcal{C}_k(\mathcal{T})$ denote the set of open k-cells of $\mathcal{T}$. The top-dimensional cells $\sigma_N \in \mathcal{C}_N(\mathcal{T})$ are termed *generalized voxels* (or bit cells).

Definition 1.2 (N-Dimensional Binary Image and Foreground Subcomplex)**.** A *binary image* on $\mathcal{T}$ is a map with compact support:

$$I : \mathcal{C}_N(\mathcal{T}) \to \{0, 1\}. \tag{1}$$

The *foreground complex* $K(I) \subseteq \mathcal{T}$ is the closed subcomplex generated by the active N-cells:

$$K(I) \coloneqq \bigcup_{\substack{\sigma_N \in \mathcal{C}_N(\mathcal{T}) \\ I(\sigma_N)=1}} \bar{\sigma}_N. \tag{2}$$

For each $k \in \{0, \ldots, N\}$, the set of active k-cells is $\mathcal{C}_k(K(I)) = \{\sigma_k \in \mathcal{C}_k(\mathcal{T}) \mid \sigma_k \subseteq K(I)\}$, and we denote $n_k(K(I)) \coloneqq |\mathcal{C}_k(K(I))|$.

Definition 1.3 (Euler-Poincaré Characteristic)**.** The Euler characteristic $\chi(K(I)) \in \mathbb{Z}$ is defined by the alternating sum of cell cardinalities:

$$\chi(K(I)) = \sum_{k=0}^{N} (-1)^k n_k(K(I)). \tag{3}$$

2 The N-Dimensional Dual Voronoi Partition

Let $\mathcal{V}(\mathcal{T}) = \mathcal{C}_0(\mathcal{T})$ denote the set of 0-cells (primal vertices) of $\mathcal{T}$.

Definition 2.1 (Dual Cell Group)**.** For each primal vertex $v \in \mathcal{V}(\mathcal{T})$, the *dual Voronoi cell* $C(v)$ is defined as:

$$C(v) \coloneqq \left\{ x \in \mathbb{M}^N \;\middle|\; d(x,v) \leq d(x,w) \quad \forall w \in \mathcal{V}(\mathcal{T}) \right\}. \tag{4}$$

The collection $\{C(v)\}_{v \in \mathcal{V}(\mathcal{T})}$ forms an N-dimensional cell complex partitioning $\mathbb{M}^N$.

Definition 2.2 (Vertex Multiplicity of k-Cells)**.** For a regular or uniform tessellation $\mathcal{T}$, each k-cell $\sigma_k \in \mathcal{C}_k(\mathcal{T})$ is the convex hull of a fixed number of boundary vertices:

$$V_k \coloneqq |\mathcal{V}(\bar{\sigma}_k)| = |\mathcal{C}_0(\bar{\sigma}_k)|. \tag{5}$$

Lemma 2.3 (Uniform Multiplicity of Cell Partitioning)**.** *Let $\{C(v)\}_{v \in \mathcal{V}(\mathcal{T})}$ be the dual cell partition of $\mathbb{M}^N$. For every $k \in \{0, 1, \ldots, N\}$ and every k-cell $\sigma_k \in \mathcal{C}_k(\mathcal{T})$:*

$$\sum_{v \in \mathcal{V}(\mathcal{T})} \mathbf{1}_{\{\sigma_k \cap \mathrm{int}(C(v)) \neq \emptyset\}} = V_k. \tag{6}$$

Proof. By vertex-transitivity and regular symmetry of $\mathcal{T}$:

1. For $k = 0$, the vertex v lies strictly at the center of $C(v)$; hence $V_0 = 1$.
2. For $k = 1$, any edge $e = (v_1, v_2)$ is transversally bisected by the common boundary interface $\partial C(v_1) \cap \partial C(v_2)$, intersecting exactly the $V_1 = 2$ dual cells.
3. For $k \geq 2$, the barycentric subdivision of σ_k divides the interior of σ_k into V_k congruent polyhedral sectors, each contained in the Voronoi cell $C(v_i)$ of one of its V_k spanning vertices $v_i \in \mathcal{V}(\bar{\sigma}_k)$. □

3 The General N-Dimensional Local Formula

Definition 3.1 (Local Neighborhood Configuration $Q(v)$)**.** For each primal vertex $v \in \mathcal{V}(\mathcal{T})$, let $\mathrm{Star}_N(v) = \{\sigma_N^{(1)}, \ldots, \sigma_N^{(D_v)}\}$ be the set of D_v distinct N-cells meeting at v. The local configuration is the binary vector:

$$Q(v) \coloneqq \left(I(\sigma_N^{(1)}), I(\sigma_N^{(2)}), \ldots, I(\sigma_N^{(D_v)}) \right) \in \{0,1\}^{D_v}. \tag{7}$$

Theorem 3.2 (N-Dimensional Local Valuation Theorem)**.** *The global Euler characteristic $\chi(K(I))$ is equal to the sum of local Euler valuations evaluated over all primal vertices:*

$$\chi(K(I)) = \sum_{v \in \mathcal{V}(\mathcal{T})} \Delta\chi(Q(v)), \tag{8}$$

where the local contribution $\Delta\chi(Q(v))$ is given by:

$$\Delta\chi(Q(v)) = \sum_{k=0}^{N} (-1)^k \Delta n_k(Q(v)), \tag{9}$$

with the localized fractional cell contributions:

$$\Delta n_k(Q(v)) \coloneqq \frac{1}{V_k} \cdot \left| \{\sigma_k \in \mathcal{C}_k(K(I)) \mid v \in \bar{\sigma}_k\} \right|. \tag{10}$$

Proof. Summing the local contributions over all vertices $v \in \mathcal{V}(\mathcal{T})$ and interchanging the order of summation:

$$\begin{aligned}\sum_{v\in\mathcal{V}(\mathcal{T})} \Delta\chi(Q(v)) &= \sum_{v\in\mathcal{V}(\mathcal{T})} \sum_{k=0}^{N} (-1)^k \Delta n_k(Q(v)) \\ &= \sum_{k=0}^{N} (-1)^k \sum_{v\in\mathcal{V}(\mathcal{T})} \frac{1}{V_k} \sum_{\substack{\sigma_k\in\mathcal{C}_k(K(I)) \\ v\in\bar{\sigma}_k}} 1 \\ &= \sum_{k=0}^{N} (-1)^k \frac{1}{V_k} \sum_{\sigma_k\in\mathcal{C}_k(K(I))} \sum_{v\in\mathcal{V}(\bar{\sigma}_k)} 1.\end{aligned}$$

By definition of V_k, the inner sum evaluates to $\sum_{v\in\mathcal{V}(\bar{\sigma}_k)} 1 = V_k$. Therefore:

$$\sum_{v\in\mathcal{V}(\mathcal{T})} \Delta\chi(Q(v)) = \sum_{k=0}^{N} (-1)^k \frac{1}{V_k} \sum_{\sigma_k\in\mathcal{C}_k(K(I))} V_k = \sum_{k=0}^{N} (-1)^k n_k(K(I)) = \chi(K(I)). \qquad \square$$

Theorem 3.3 (Spherical Link Formulation)**.** *Let $S_\varepsilon^{N-1}(v)$ be a geodesic sphere of sufficiently small radius $\varepsilon > 0$ centered at $v \in \mathcal{V}(\mathcal{T})$, and let $\mathrm{Lk}(v, K(I)) \coloneqq K(I) \cap S_\varepsilon^{N-1}(v)$ denote the lower/spherical link. For connected vertex topologies, the local valuation satisfies:*

$$\Delta\chi(Q(v)) = \begin{cases} 1 - \chi(\mathrm{Lk}(v, K(I))) & \text{if } Q(v) \neq \mathbf{0}, \\ 0 & \text{if } Q(v) = \mathbf{0}. \end{cases} \tag{11}$$

4 Specialization to 3-Dimensional Volume Processing ($\mathbb{Z}^3$)

Let $\mathcal{T} = \mathbb{Z}^3$ be the standard 3D cubic lattice. Each voxel is a closed unit cube $\sigma_3 = [x, x+1] \times [y, y+1] \times [z, z+1]$.

4.1 Combinatorial Parameters for $\mathbb{Z}^3$

- $V_0 = 1$ (1 vertex per 0-cell).
- $V_1 = 2$ (2 vertices per 1-edge).
- $V_2 = 4$ (4 vertices per 2-square face).
- $V_3 = 8$ (8 vertices per 3-cube voxel).
- At every grid intersection $v \in \mathbb{Z}^3$, exactly $D_v = 2^3 = 8$ voxels meet.

The local configuration $Q(v) \in \{0,1\}^{2\times2\times2}$ is a $2 \times 2 \times 2$ bit octocube ($2^8 = 256$ possible local binary patterns).

4.2 The 3D Local Contribution Formula

For any bit octocube $Q(v) \in \{0,1\}^{2\times2\times2}$:

$$\Delta\chi(Q(v)) = \Delta n_0(Q) - \Delta n_1(Q) + \Delta n_2(Q) - \Delta n_3(Q), \tag{12}$$

where:

$$\Delta n_3(Q) = \frac{1}{8}\sum_{i=1}^{8} I(\sigma_3^{(i)}), \tag{13}$$

$$\Delta n_2(Q) = \frac{1}{4}\cdot\left|\{\text{active square faces incident to } v \text{ in } Q\}\right|, \tag{14}$$

$$\Delta n_1(Q) = \frac{1}{2}\cdot\left|\{\text{active unit edges incident to } v \text{ in } Q\}\right|, \tag{15}$$

$$\Delta n_0(Q) = \begin{cases} 1 & \text{if } Q \neq \mathbf{0} \text{ (under 26-connectivity / 1s-joined)}, \\ 0 & \text{if } Q = \mathbf{0}. \end{cases} \tag{16}$$

Multiplying by the common denominator $\mathrm{lcm}(V_0, V_1, V_2, V_3) = 8$ gives the integer-normalized equation:

$$8\chi(K(I)) = \sum_{v\in\mathbb{Z}^3} \left(8\Delta n_0(Q(v)) - 4\cdot n_1^{\mathrm{loc}}(Q(v)) + 2\cdot n_2^{\mathrm{loc}}(Q(v)) - n_3^{\mathrm{loc}}(Q(v))\right). \tag{17}$$

Under the full octahedral symmetry group O_h of the cube, the 256 possible octocube configurations partition into exactly **22 equivalence classes**.

5 Dimensional Scaling on Hypercubic Lattices $\mathbb{Z}^N$

Table 1: Valuation parameters for hypercubic lattices $\mathbb{Z}^N$ across dimensions.

Dimension N	**Window Configuration**	**Local State Space**	**Denominator** $\mathrm{lcm}(V_k)$	**Order** R	**D**
1D	1×2 array	$2^2 = 4$	2	$R = 2$	
2D	2×2 bit quad	$2^4 = 16$	4	$R = 4$	
3D	$2 \times 2 \times 2$ octocube	$2^8 = 256$	8	$R = 8$	
ND	2^N hypercube	2^{2^N}	2^N	$R = 2^N$	

Corollary 5.1 (Universal Hypercubic Denominator)**.** *On the standard hypercubic lattice $\mathbb{Z}^N$, the vertex multiplicities satisfy $V_k = 2^k$. Consequently, the natural normalization factor clearing all fractional denominators is:*

$$\mathrm{lcm}(V_0, V_1, \ldots, V_N) = 2^N. \tag{18}$$